

SegWrap

Problem: Traditional video object segmentation struggles to maintain temporal consistency.

Approach: We use diffusion models to iteratively refine segmentation masks from input iframes.

Key contribution: A two-stage inference pipeline for I-frames and B-frames that preserves temporal consistency and mask quality.

CoVOS Foundation: I-Frames & B-Frames

Core Insight: Video codecs already compute motion. CoVOS exploits this to avoid redundant computation.

I-Frame (Keyframe)

- Complete image
- $P_i = \text{SegDiff}(x_T, F_i)$, $x_T \sim \mathcal{N}(0, I)$
- Output: Pristine initial mask from full 100-step reverse diffusion

B-Frame (Predictive)

- Stores only motion vectors (move pixels left/right)
- Predictor: Warp mask using free motion vectors:
 $\hat{P}_i = \text{Warp}(P_{i-1}, MV_{i-1 \rightarrow i})$
- Corrector: Light model snaps to true boundaries:
 $P_i = \theta_{\text{light}}(\hat{P}_i, F_i)$

Sequence & Output

$$[F_0^{\text{I-Frame}} \mid F_1, F_2, \dots \mid F_k^{\text{I-Frame}} \mid \dots] \longrightarrow V_{\text{seg}} = [P_0, P_1, P_2, \dots, P_N]$$

SegDiff Architecture

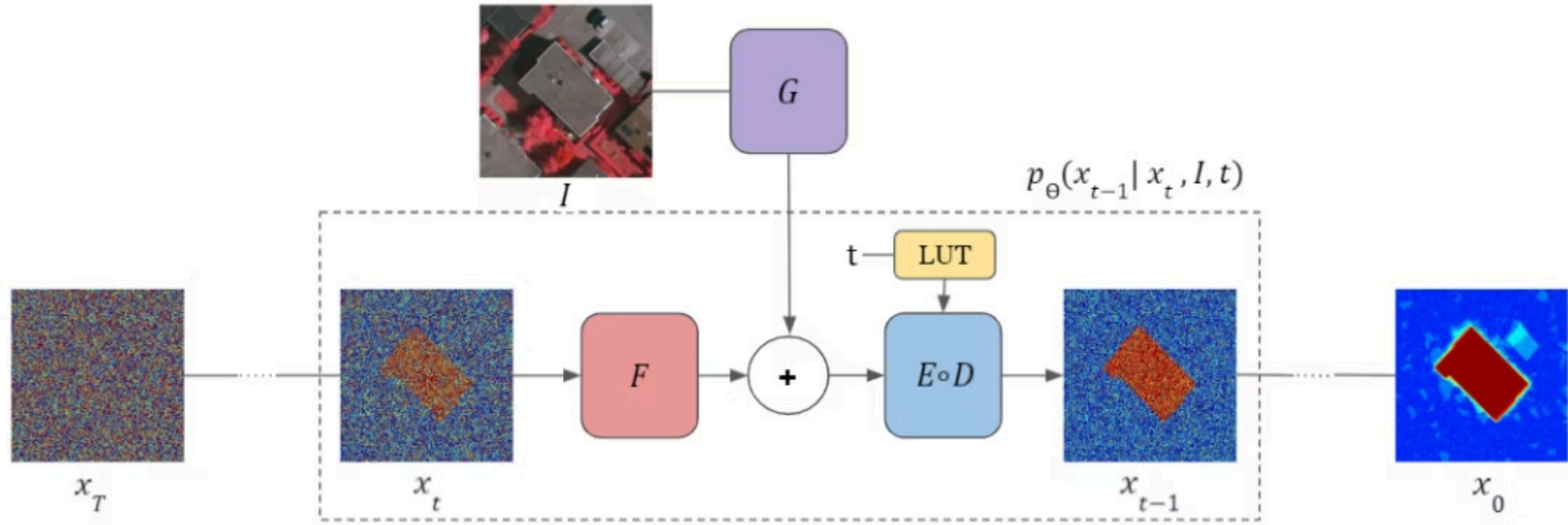


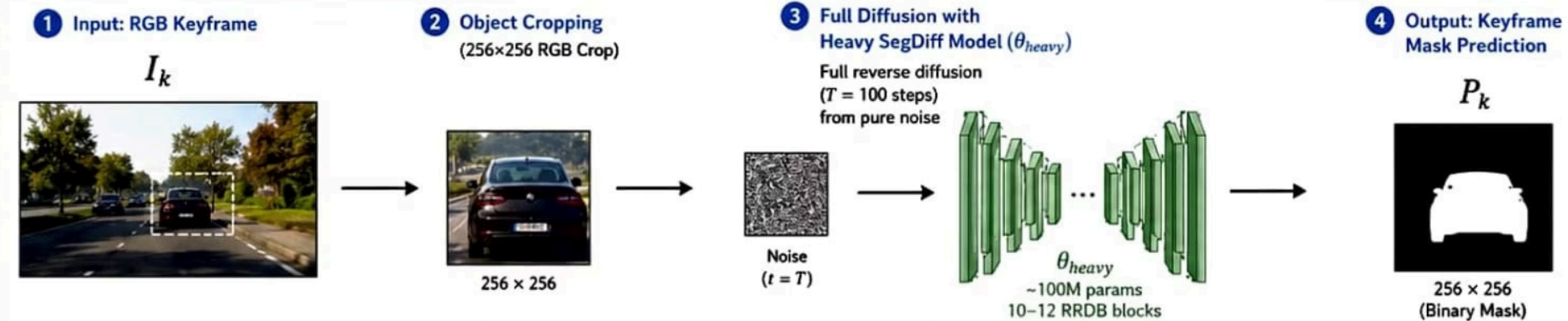
Figure 1. Our proposed diffusion method for image segmentation encodes the input signal, x_t , with F . The extracted features are summed with the feature map of the conditioned image I generated by network G . Networks E and D are a U-net encoder and decoder [35, 38], respectively, that refine the estimated segmentation map, obtaining x_{t-1} .

SegWrap Architecture

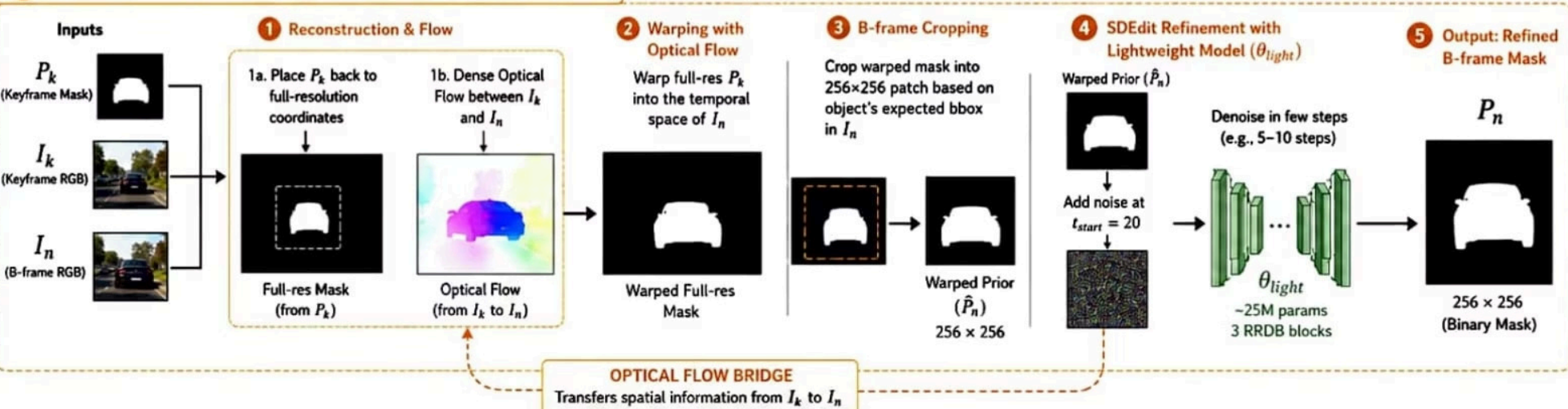
SegWrap is a segmentation framework that combines a shared feature encoder with task-specific wrapping and prediction modules. It operates on multiscale visual features to produce spatially consistent mask outputs while preserving boundary detail.

Diffusion Model-Based VOS with Keyframe (I-frame) and Non-Keyframe (B-frame) Pipeline

1 I-FRAME (KEYFRAME) PIPELINE – HEAVY PIPELINE



2 B-FRAME (NON-KEYFRAME) PIPELINE – LIGHTWEIGHT REFINEMENT



Initial Dataset Exploration

Vaihingen & MoNuSeg

Static image datasets,
but no video context.

Cityscapes

Missing key annotations,
and the authors were
unreachable.

ADE20K

Too many classes for our
small model capacity.

Decision

Switched to video
datasets (YTVOS →
YTVIS) for I-frame
training.

Dataset Transition: YTVOS → YTVIS

Problem: YTVOS annotation inconsistency (multiple vs. single masks per frame) degraded model performance.

Solution: Switched to YTVIS with consistent single-instance annotations and smart cropping.

YTVOS Results

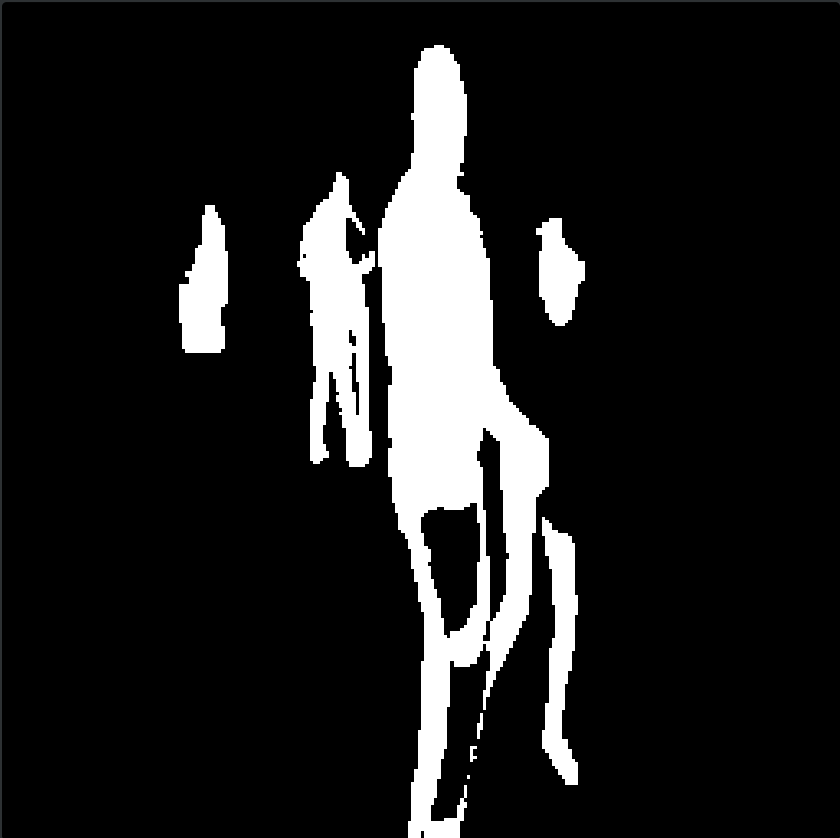
Metric	Value
IoU	0.468
F1	0.574
WCOV	0.631
F-Bound	0.501

Results

Input Frame



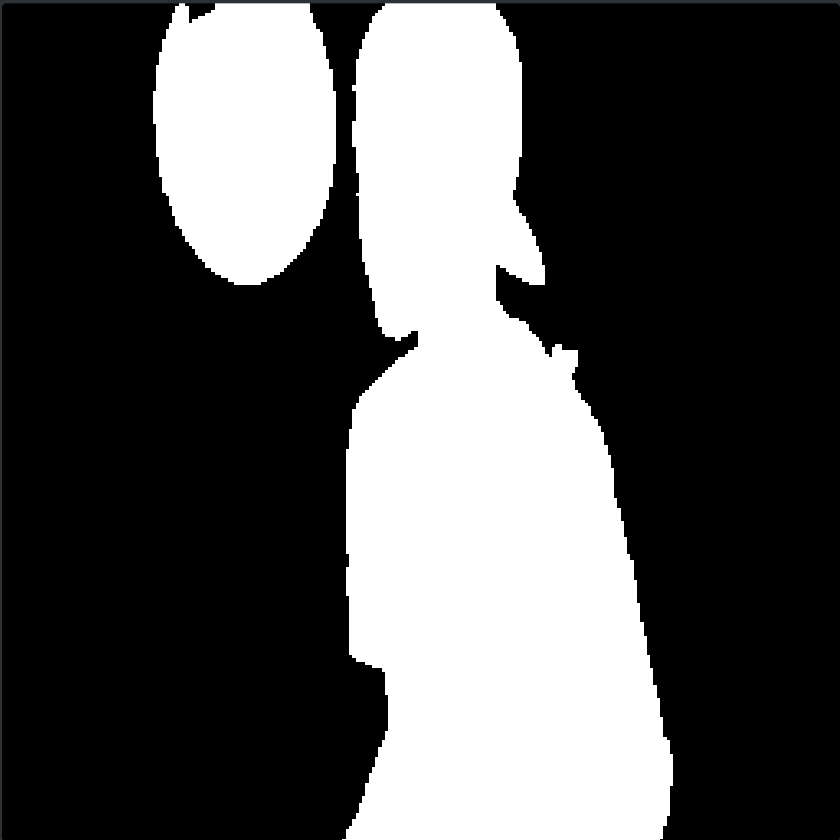
Segmentation Output



Input Frame



Segmentation Output



Results: YTVIS Benchmark

0.734

Mean IoU

0.831

Mean F1

Training :150k iterations on YTVIS Dataset, finetuned for 20k iterations on DAVIS 16
Inference time: 100 seconds per iframe AND 20 seconds per bframe on L40s (majority vote number = 10)

B-Frame Comparison

Case	Warp-only IoU	Warp-only F1
GT	0.615	0.704
I-Frame Pred	0.503	0.623

Technical Innovations

1

Computational Efficiency

Heavy model inference is run only on I-frames, while B-frames use lightweight warping and correction to reduce compute cost.

2

Occlusion Handling

Periodic I-frame resets rediscover occluded objects and prevent permanent tracking failure.